

Spatial Resolution Tier Boundary Validation in Hierarchical Hexagonal Biosphere Simulation Matrices: Sprint 023 Report

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Abstract

Computational ecosystem models require rigorous spatial discretization to simulate biogeochemical cycles and energy flows across planetary surfaces. The Web of Life simulation matrix utilizes Uber’s H3 hierarchical hexagonal spatial indexing system, spanning 16 discrete resolution tiers ($r \in [0, 15]$). Sprint 023 introduces a strict boundary check module within `src/spatial/h3_grid.ts` and integrates it into the `SpatialMonad` architecture. By enforcing formal tier validation predicates, the simulation prevents out-of-bounds indexing errors, ensures topological manifold consistency, and maintains strict adherence to the First and Second Laws of Thermodynamics regarding matter conservation and state-space bounding during spatial stock transitions.

1 Introduction and Thermodynamic Context

The Web of Life framework models the biosphere as a complex thermodynamic system governed by solar input, carbon cycling, and matter conservation. Spatial partitioning via Uber’s H3 hexagonal grid facilitates multi-scale ecological simulation. However, abstract spatial indexing must adhere to physical and geometric limits defined by the geosphere’s partitioning constraints.

- **First Law of Thermodynamics:** Spatial indexing operations do not consume or generate matter or energy ($\Delta M = 0, \Delta E = 0$). Refinement operations must preserve underlying ecological stock vectors.
- **Second Law of Thermodynamics:** Entropy and information constraints dictate that resolution parameters must be bounded integers, preventing infinite-precision singularities and state corruption within monad transitions.

2 Mathematical Formulation

Let the spatial monad state $\mathcal{S}$ encapsulate a stock vector $\mathbf{x} = [C, H_2O, Min, O_2, E]^T$ at an H3 index with resolution r . The validation predicate $\Pi(r)$ is defined as:

$$\Pi(r) = \begin{cases} 1 & \text{if } r \in \mathbb{Z} \text{ and } 0 \leq r \leq 15 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

If $\Pi(r) = 0$, a `RangeError` is thrown, halting monad transitions and preserving existing stock inventories.

3 Implementation in TypeScript

The core resolution validation functions added in `src/spatial/h3_grid.ts` are defined as follows:

```
import { Resolution } from './h3_types';

export const MIN_H3_RESOLUTION: Resolution = 0;
export const MAX_H3_RESOLUTION: Resolution = 15;

export function isValidH3Resolution(resolution: number): boolean {
  return Number.isInteger(resolution) && resolution >=
    MIN_H3_RESOLUTION && resolution <= MAX_H3_RESOLUTION;
}

export function assertH3Resolution(resolution: number): void {
  if (!isValidH3Resolution(resolution)) {
    throw new RangeError(`Invalid H3 resolution tier: ${resolution}.
      Must be an integer between ${MIN_H3_RESOLUTION} and ${
        MAX_H3_RESOLUTION}.`);
  }
}
```

4 Verification and Compliance Matrix

Empirical verification via `tests/sprint_023.test.ts` confirms compliance across all test vectors:

1. **TC-01:** Integers 0 through 15 successfully pass validation.
2. **TC-02 & TC-03:** Negative inputs ($r < 0$) and excessive inputs ($r > 15$) correctly trigger `RangeError`.
3. **TC-04:** Non-integer floating-point tiers (e.g., $r = 7.5$) are rejected.
4. **TC-05:** Spatial monad refinement maintains exact conservation invariants ($\Delta M = 0$, $\Delta E = 0$).

5 Conclusion

Sprint 023 establishes a foundational geometric safeguard within the Web of Life spatial indexing engine. By integrating rigorous boundary checks into both grid utilities and monadic state pipelines, the simulation matrix preserves thermodynamic integrity across all hierarchical scales.

Project Repository: <https://github.com/pascalranoroarijaona/WebOfLife>